

Somatic Continuum: A Body-Controlled, Generative Csound Instrument

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Fig. 1. The visual layer of the installation: the visitor's face and hands rendered live as flowing lines in WebGL2. Still from the installation, Visceral Design, 2026.

Abstract. Somatic Continuum is an interactive installation in which a visitor's face, hands, heartbeat, and presence become the instrument. There are no buttons and nothing to hold. A camera reads the face through MediaPipe FaceMesh, a hand tracker reads the hands and the movement of the arms, and a worn heart monitor reads the pulse. All of it drives a live Csound orchestra of more than twenty instruments over Open Sound Control. The instruments are spread across nine worlds, and each world puts a different instrument on each hand. A shared key keeps every combination in tune. The visitor's heartbeat pulses the image and sets the timing of the arpeggios, so image and sound move with the body. Each person who steps in and connects their heartbeat opens a new sonic world. Nothing is recorded in advance. The music is generated live by rules that the body shapes as it moves. Koenig and Tenney composed by setting rules and letting them generate the music; Somatic Continuum does the same, except the rules run in real time and the body sets them in motion. This installation gives Csound a new kind of interface, one played with the body, that lets an untrained visitor compose.

Keywords: Csound, Open Sound Control, web-based Csound, generative music, algorithmic composition, somatic interaction, science art, interactive installation, real time synthesis

1. Introduction

An instrument usually stands between the body and the sound. Somatic Continuum removes it. A visitor's face, hands, heartbeat, and presence are the whole interface to a live Csound engine, with no controller to hold and no sheet music to follow. This installation asks a question. What is left of an instrument when the body plays it directly, and what kind of music lets someone with no training sound expressive rather than lost?

Somatic Continuum reads facial landmarks, full hand skeletons, and the participant's heart, and turns them into generative visuals and live sound. The body signals travel over Open Sound Control to a local Csound orchestra. Two ideas shape the work. First, Csound runs as its own separate program, so the audio never has to share the processor with the face tracking or the graphics. That is what keeps the sound stable. Second, the music is generated by rules that change with each interaction rather than written down. The instruments carry their own logic and a shared harmonic field.

2. System Architecture

Somatic Continuum has four layers on one machine (Fig. 2). A perception layer produces the body signals. A graphics layer produces the visuals. An OSC bridge carries the control data. Csound produces the sound.

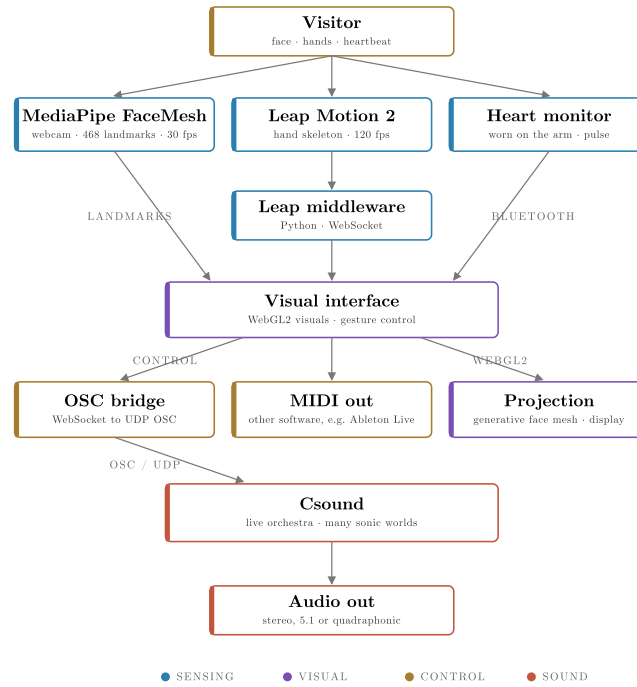


Fig. 2. System signal flow, from the body to Csound and out to sound.

Perception. The perception layer uses MediaPipe FaceMesh [1] with landmark refinement, which gives 468 facial landmarks and iris tracking at about 30 frames per second from a camera. From these it derives mouth opening, eye state, brow height, and head tilt. An Ultraleap Leap Motion Controller 2 [5] tracks the hands at up to 120 frames per second [11], about four times the rate of the face tracker. A small Python program passes the hand data to the visual interface. A worn heart monitor adds a third, involuntary input: the participant's pulse and the way it varies. The camera delivers a new set of landmarks every 33 ms and the hand tracker reports within 15 ms [11]. The bridge sends OSC on the local machine, and Csound runs at 44.1 kHz with a control period of 64 samples, 1.5 ms. Each incoming control value therefore reaches the sound within a few milliseconds.

Visuals. The visuals are a multi-pass renderer in WebGL2 [4]. It draws the face as flowing perforated lines with particle trails from the hands, adds a soft bloom, and offers normal, kaleidoscope, prism, and mirror visual worlds. The heartbeat pulses through the image, so the visuals breathe with the body (Figs. 1 and 3).

Audio over OSC. A small bridge turns the interface's messages into OSC packets [3], which Csound [2] receives with its **OSClisten** opcode. Csound then synthesizes many instruments and effects including drones, synths, chant, bowls, bass, chords and more. The same control data is also sent out as MIDI, so other software such as Ableton Live can be played, mixed and layered with Csound. The same design has also been run on GNU/Linux with PipeWire [6]. Running Csound as its own program, apart from the face tracking and the graphics, is the central decision. It keeps the audio stable and opens the piece to the full depth of the Csound opcode set and the software synthesis tradition documented in The Csound Book [9].

3. Csound Instrument Design

The Csound orchestra holds more than twenty instruments, and those instruments are built rather than sampled. The choir and the Tibetan chant use FOF formant synthesis [10], which shapes the vowel resonances of a human voice. The bowed strings are a **wgbow** physical model. The sitar drone and tanpura underpin the harmony. They layer detuned **vco2** oscillators through **streson** resonators that ring like sympathetic strings, then a **moogladder** filter and a long reverberant tail. Head tilt bends the drone and changes the panning. The Rhodes and the electric piano come from **foscil** FM. A lush pad, analog brass, and a glass crystal pad are drawn from detuned **vco2** oscillators through a **moogladder** filter. Around these voices sit a mouth pad, singing bowls, an 808 kick, a Reese bass, and a chord pad, and each voice keeps its own **reverb**sc reverberation. A trip hop beat anchors one of the worlds and runs through custom Csound effects chain. The same gesture sounds like a bowed string in one world and a Rhodes keyboard in the next. A shared key holds the whole orchestra in tune, and a master filter under a pinch shapes the sound.

4. Generative Process and Harmonic Design

The music is generated live by rules, not written note by note. Each instrument follows its own logic. The drone moves through the scale on its own timer, across many movement patterns, with no outside clock. The arpeggiator takes its timing from the pulse: each step lasts 15000 milliseconds divided by the heart rate in beats per minute, held to no faster than 110 ms, so at 60 beats per minute a step lasts 250 ms and four notes fall inside every heartbeat. Without a pulse reading, it runs at 190 ms. The arpeggio cycles through the notes of the current chord, drawn from a chord list in the shared key, and the drone steps through a fixed list of base notes in that key while head tilt adds up to 25 Hz of bend. The harmony stays inside a shared field, so layers that evolve on their own still sonically match. The visitor plays without reading music or knowing which note is which. A movement of the hand chooses a pitch, and the running processes carry the rest, so the result stays inside a consonant harmonic field whatever the visitor does.

Koenig built composing programs that let rules generate a score [7]. Tenney shaped music through formal and stochastic processes rather than choosing each note [8]. Somatic Continuum carries that idea with the embodied Csound instruments, where a visitor's body, not a program, sets the rules in motion. The self evolving synthesis inside this interactive installation allows a person to make art with their own body. This installation creates a generative field which allows untrained skill and mysterious exploration into sonic and artistic intentionality.

The shared key is a deliberate trade. What the visitor determines is real: the pitch region each arm reaches, the filter and its cutoff through the pinch of the fingers, when the mouth sets a sound off and when closed eyes bring new patterns, the bend of the drone through head tilt, and the timing through the pulse. What is guaranteed in advance is the harmonic frame. Every combination stays consonant, so no gesture can sound wrong, and the harmony holds its centre rather than travelling. Nothing can sound wrong, but nothing happens without engagement: the piece is nothing until someone steps in, and the more the visitor engages, the richer the soundscape becomes and the more the instrument can do. One key across all the worlds is the deliberate price of that guarantee. This differs from Koenig, whose programs were built to carry him past his own preferences [7]. Here the rules hold the visitor inside the frame rather than push against the author. The visitor composes surface, texture, timing, and colour inside a frame the author has fixed.

5. Body-to-Sound Mappings

Each part of the body drives a distinct layer, so visitors learn the piece by trying it. Each arm plays a different instrument, and raising or lowering the arms blends them in harmony, while the fingers unlock filters and settings inside each one. Opening the mouth can call a chant, start an arpeggiator, or change the filters and resonances of the instruments already being heard. Closing the eyes can bring up new arpeggiators and patterns, with their own effects, all made in Csound. Tilting the head bends the drone, and turning it moves the sound in space. Raised brows brighten a resonance. Bringing the hands together triggers a different change in each world. A visitor can also record their own voice and layer it into the piece. Through all of it runs the heartbeat, which pulses the image, shifts its color, and sets the timing of the arpeggios and delays, so sound and image stay in synch with the body. Because every layer shares one key, visitors play freely without falling out of key. They are also able to take pictures and video so that they can keep a part of the art that they created.



Fig. 3. A visitor's face and hands driving the visual and the Csound audio, pulsed by the heartbeat. Still from the installation, Visceral Design, 2026.

6. Development and Discussion

The instrument reached its form through a long, hand tuned process, and its most important decision came from failure. In its first iterations Somatic Continuum synthesized its sound inside the same program that ran the face tracking and the graphics. Under load, with a visitor moving quickly in front of the camera, the audio glitched and dropped out: one process could not carry landmark detection, a WebGL2 particle renderer and real-time synthesis at once, and the sound was the first thing to suffer. The sound also could not be as rich and layered as intended in those first iterations. In that form the installation was not possible. Moving the sound into a standalone Csound process changed the piece. The interface now sends only control data over OSC, and the audio no longer depends on what the graphics or the tracker are doing. The difference is structural. In the earlier build one process carried the face tracker at about 30 frames per second, the WebGL2 renderer and the audio callback, and the audio broke up whenever that process fell behind. In the current build Csound is a separate process running at 44.1 kHz with a 64-sample control period (1.5 ms) and a 512-sample software buffer (11.6 ms), so a slow frame on the graphics side can no longer starve the audio. The dropouts stopped, the instruments could be designed in the orchestra file with the full opcode set, from the FOF voices and the **wgbow** model to the **streson** resonators and per-voice **reverb**, and the same control stream could be sent onward as MIDI to other software.

A second lesson came from gesture handling. Fast hand movement used to produce a churn of retriggered notes, and the sound broke up at exactly the moment the visitor was most expressive. Holding the current chord and bending its pitch during fast motion, then releasing it when the hand settles, removed the break-up and made fast gestures feel like playing rather than triggering. Visitors at public showings, including the Linux Audio Conference 2026 in Maynooth and NIME in London, responded first to the feeling of making their own music. Most asked to keep a picture of their face in the mesh, which the piece offers, and in that sense the concept succeeded: people felt they went home with their own piece of art, created by them. Newcomers often needed the piece explained and demonstrated to be able to get full use of all the features.

7. Conclusions

Somatic Continuum shows how face tracking, hand tracking, a heart monitor, OSC, and a carefully built Csound orchestra can make one generative instrument that untrained visitors play into an interactive art piece that they feel emotionally connected to. It brings computer vision, physiology, and live synthesis into a single Csound installation, with rule based synthesis at its centre, and it carries forward the way Koenig and Tenney composed, by setting rules and letting those rules generate the music. A demonstration video is available at <https://drive.google.com/drive/folders/1dNEBjEfsQBY9SEowypCW957QcFErhOnm?usp=sharing>

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